

MILL / MACHINE

DATE

OPERATOR

TARGET GSM

1 · PREP & SOURCING (TICK WHEN IN HAND)

2 · BLEND RECIPE (% OF OVEN-DRY FURNISH)

| ✓                        | MATERIAL                                   | INDICATIVE ₹ (CONFIRM BY QUOTE) | COMPONENT               | T1<br>DROP-IN | T2<br>+BAMBOO | T3<br>TUNE |
|--------------------------|--------------------------------------------|---------------------------------|-------------------------|---------------|---------------|------------|
| <input type="checkbox"/> | Domestic OCC (box scrap)                   | ₹14–23/kg                       | Domestic OCC            | 70            | 60            |            |
| <input type="checkbox"/> | Jute mill waste / old gunny (NOT raw jute) | gunny scrap ~₹15–17/pc          | Imported fibre          | 0             | 0             | 0          |
| <input type="checkbox"/> | Domestic kraft trim / DLK                  | ₹18–24/kg                       | Bagasse                 | –             | 10            |            |
| <input type="checkbox"/> | Bagasse (depithed)                         | ₹1.4–2.2/kg                     | Jute waste / gunny      | 15            | 10            |            |
| <input type="checkbox"/> | Bamboo unbleached pulp (premium)           | ₹58–62/kg                       | Domestic kraft trim     | 15            | 10            |            |
| <input type="checkbox"/> | Cationic starch                            | ₹45–70/kg                       | Bamboo pulp             | –             | 10            |            |
| <input type="checkbox"/> | GPAM + dry-strength resin/PVAm (trial kit) | quote                           | Cationic starch         | 1.0%          | 1.0%          |            |
| <input type="checkbox"/> | Refiner enzyme (cellulase blend)           | quote                           | GPAM                    | 0.2%          | 0.2%          |            |
|                          |                                            |                                 | Dry-strength resin/PVAm | 0.3%          | 0.3%          |            |
|                          |                                            |                                 | Enzyme (at refiner)     | 0.3kg/t       | 0.3kg/t       |            |

Suppliers: starch — Gujarat Ambuja / Sukhjit / Roquette · GPAM/DSR — Anmol Polymers, UP Alums, Solenis/Kemira/Ecolab/Buckman · enzyme — Advanced Enzyme Technologies, Novonesis.

If a target is missed: starch 0.5–1.5% · GPAM 0.1–0.3% · resin 0.2–0.5% · enzyme 0.1–0.5 kg/t · long-fibre 10 → 25%.

3 · TESTS & PASS TARGETS VS CONTROL

| PROPERTY          | METHOD                                   | INSTRUMENT         | PASS TARGET (VS CONTROL)   | T1 ✓/<br>✗ | T2 ✓/<br>✗ | T3 ✓/<br>✗ |
|-------------------|------------------------------------------|--------------------|----------------------------|------------|------------|------------|
| GSM               | ISO 536                                  | Balance            | Grade ±2%                  |            |            |            |
| BF (Burst Factor) | IS 1060 / ISO 2758 / T403                | Mullen burst       | Within ±1 BF point         |            |            |            |
| RCT (Ring Crush)  | ISO 12192 / TAPPI T822                   | Ring-crush         | ≥ 95% of control           |            |            |            |
| CMT (if fluting)  | ISO 7263 / T809                          | Crush tester       | ≥ 95% of control           |            |            |            |
| Cobb (water abs.) | ISO 535                                  | Cobb tester        | Within ±10%                |            |            |            |
| Colour / shade    | ΔE or light-box                          | Spectro / daylight | Visually matched           |            |            |            |
| Smell             | 5-person blind panel, fresh + 24h bagged | —                  | Neutral / no off-odour     |            |            |            |
| Moisture          | ISO 287                                  | Oven / meter       | Within spec                |            |            |            |
| Runnability       | Drainage °SR/CSF, breaks, speed          | Machine DCS        | Speed ≥ 95%, no new breaks |            |            |            |

4 · RESULT & COST

| RECORD                                         | CONTROL   | T1 | T2 | T3 |
|------------------------------------------------|-----------|----|----|----|
| BF / RCT (N)                                   |           |    |    |    |
| Colour / Smell (Y-N)                           | ref / ref |    |    |    |
| Machine speed                                  |           |    |    |    |
| Furnish ₹/tonne = Σ(% × ₹/kg × 10) + chemistry |           |    |    |    |
| % imports used                                 |           | 0% | 0% | 0% |
| ₹ saved / tonne                                | —         |    |    |    |

**DECISION RULE:** the cheapest blend that PASSES all of BF, RCT, colour, smell and runnability is the product. **Watch-items:** jute/gunny & agro can affect **smell** and **drainage** — those two lines decide acceptance as much as strength. All ₹ figures are indicative public-listing ranges (India 2025) — **confirm every price by quotation.** Bamboo is legal (not a "tree" since the 2017 Forest Act amendment).